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# Adsorption performance of radial-flow contactors for direct air capture: A computational study

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ABSTRACT

The need for efficient direct air capture (DAC) is increasing as the global mean temperature continues to rise and a large fraction of $CO_2$ emissions cannot be addressed by conventional $CO_2$ capture technologies. For adsorption-based DAC, the geometry of the gas-solid contactor is an important design parameter because it determines pressure losses, sorbent utilisation, and the efficiency of the process. This study investigates the adsorption stage in a centripetal Z-type radial-flow adsorber for DAC. A computational fluid dynamics (CFD) model is adopted to simulate the gas flow through the porous bed of adsorbent particles adsorbing $CO_2$ from the gas with the help of the linear driving force model and the Toth isotherms. The material properties are experimentally obtained for a commercial amine-functionalised resin. Twelve different configurations of the contactor are analysed with various bed aspect ratios and radius ratios while keeping the bed volume constant. The simulations indicate that the geometry strongly affects the flow distribution, pressure drop, and adsorption time, which affect the specific energy requirement and productivity of the adsorption stage of the DAC process. Configuration selection represents the trade-off between low energy requirements (at high bed aspect ratio and radius ratio), low space requirements, and construction material costs (at low bed aspect ratio and radius ratio). The intermediate geometries provide the most practical basis for scaling up to a 7-contactor adsorber.

## 1. Introduction

Meeting the Paris Agreement's long-term global temperature goals requires rapid development of carbon dioxide removal (CDR) technologies in addition to the reduction of greenhouse gas emissions (Smith et al., 2024). The reason is that half of $CO_2$ emissions come from smaller and distributed sources, e.g. in transportation, and they cannot be reduced by conventional $CO_2$ capture methods (IPCC, 2022). Direct extraction of $CO_2$ from ambient air (direct air capture, DAC) with subsequent geological storage of $CO_2$ is a promising technology for CDR that has received increasing attention in recent years (IEA, 2023). It offers a high mitigation potential and durable storage of $CO_2$ (Smith et al., 2024). However, DAC remains among the most expensive and energy-intensive CDR approaches currently available, making improvements in process efficiency and equipment design across all stages of the process critical for its implementation (Smith et al., 2024). DAC technologies mainly include adsorption-based (solid DAC) and absorption-based (liquid DAC) approaches, with amine-based materials being widely investigated for capture from dilute $CO_2$ streams (Hamdy et al., 2021). More broadly, emerging $CO_2$ capture approaches include membrane, cryogenic, and electrochemical technologies, which continue to be developed towards larger-scale implementation (Hekmatmehr et al., 2024; Fatoorehchi et al., 2026).

This work focuses on adsorption-based DAC that uses solid sorbents to bind $CO_2$ from the air. Solid adsorbents used in DAC can be categorised based on their interaction with $CO_2$ into physical sorbents, e.g. activated carbon, zeolites, metal-organic frameworks (MOFs), and chemical sorbents, such as solid alkali sorbents and amine-modified sorbents (Jiang et al., 2023). Chemical sorption forms strong chemical bonds, while physical sorption relies on weaker physical interactions (Shi et al., 2020). Physical sorbents have lower binding energy, resulting in lower $CO_2$ selectivity and capacity, while chemical sorbents are more effective at capturing $CO_2$, especially from low atmospheric concentrations. However, chemical sorbent regeneration is energetically costly compared to physical sorbents due to the stronger bonds (Abdullatif et al., 2023). Amine-based chemical sorbents are among the most widely studied adsorbents for DAC (Jiang et al., 2023; Wang et al., 2022). To improve $CO_2$ capture efficiency while reducing regeneration

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**Nomenclature**

*Abbreviations*

| | |
|---|---|
| AR | Aspect ratio |
| BET | Brunauer–Emmett–Teller |
| CDR | Carbon dioxide removal |
| CFD | Computational fluid dynamics |
| DAC | Direct air capture |
| FTIR | Fourier-transform infrared spectroscopy |
| LDF | Linear driving force |
| MOF | Metal-organic framework |
| RR | Radius ratio |
| SEM | Scanning electron microscopy |
| SER | Specific energy requirement |

*Latin symbols*

| | |
|---|---|
| $A_{fs}$ | fluid-solid interfacial area density, $m^2/m^3$ |
| AR | bed aspect ratio |
| $b$ | affinity coefficient in the Toth model, 1/Pa |
| $C_2$ | inertial resistance, 1/m |
| $c_p$ | specific heat capacity, J/(kg·K) |
| $D_m$ | mass diffusivity, $m^2/s$ |
| $d$ | diameter, m |
| $E$ | specific total energy, J/kg |
| $e$ | thermal effusivity, $Ws^{1/2}/m^2K$ |
| $\Delta H$ | adsorption enthalpy, J/mol |
| $h$ | specific enthalpy, J/kg |
| $h_{fs}$ | fluid-solid heat transfer coefficient, $W/(m^2K)$ |
| $J$ | diffusion flux, $kg/m^2s$ |
| $k$ | thermal conductivity, W/m·K |
| $k_{LDF}$ | mass transfer coefficient in the LDF model, 1/s |
| $L$ | length, m |
| $M$ | molar mass, kg/mol |
| $m$ | mass, kg |
| $\dot{m}$ | mass flow rate, kg/s |
| $n$ | amount of substance, mol |
| $\dot{n}_{tot}$ | total molar flow rate, mol/s |
| $p$ | pressure, Pa |
| $p_{CO_2}$ | partial pressure of $CO_2$, Pa |
| $\Delta p$ | pressure drop, Pa |
| $q$ | adsorption capacity, mol/kg |
| $q_m$ | maximum adsorption capacity in the Toth model, mol/kg |
| RR | bed radius ratio |
| $R$ | outer radius, m |
| $r$ | inner radius, m |
| $S_{CO2}$ | $CO_2$ mass sink, $kg/m^3s$ |
| $S_E$ | energy sink, $J/m^3s$ |
| SER | specific energy requirement, J/kg |
| $T$ | temperature, K |
| $t$ | time, s |
| $t_{Toth}$ | heterogeneity parameter in the Toth model |
| $V_{cell}$ | volume of a computational grid cell, $m^3$ |
| $v$ | velocity, m/s |
| $v_\infty$ | superficial velocity, m/s |
| $Y$ | mass fraction |
| $y$ | molar fraction |

*Greek symbols*

| | |
|---|---|
| $\alpha$ | permeability, $m^2$ |
| $\varepsilon$ | porosity |
| $\eta$ | flow uniformity coefficient |
| $\mu$ | dynamic viscosity, Pa·s |
| $\rho$ | density, $kg/m^3$ |

*Subscripts*

| | |
|---|---|
| ads | adsorbent |
| $CO_2$ | carbon dioxide |
| eq | equilibrium |
| f | fluid |
| in | inlet |
| out | outlet |
| p | particle |
| ref | reference |
| s | solid |
| sat | saturation |
| 0 | initial |
| 1 | final |

*Dimensionless numbers*

| | |
|---|---|
| Pr | Prandtl number |
| Re | Reynolds number |
| Nu | Nusselt number |

energy costs, amines, such as polyethylenimine, are functionalised on porous solid supports, such as silica, alumina, MOFs, or polymeric resins (Jiang et al., 2023; Shi et al., 2020; Sanz-Pérez et al., 2016).

A well-studied configuration of gas-solid contactors for DAC is the fixed-bed column due to its simple and process-efficient layout. In a fixed-bed contactor, the sorbent stays stationary in a packed bed while the gas flows through it Zhu et al. (2022). Fixed beds can experience a large pressure drop that limits the particle sizes used and the flow rates applied; therefore, other fixed-bed configurations are suggested instead of a column aimed to increase the contact surface area and reduce the pressure drop. Such fixed bed configurations include, for example, radial-flow annular beds, flat ""pancake-shaped" beds, and structured beds (Zhu et al., 2022).

In a radial-flow adsorber, the sorbent is packed in a coaxial cylindrical annulus separating the central tube from the outer one, where the gas flows radially between these concentric chambers through the bed (Yu and Brilman, 2020). Radial-flow contactors are classified into centripetal when the gas flows from the outer annulus toward the centre, and centrifugal when it flows from the centre outward (Xu et al., 2022b). Based on the axial arrangement of the inlet and outlet, a radial-flow contactor can be Z-type when the inlet and outlet are located at opposite ends or a π-type when they are on the same end (Xu et al., 2022b). In addition to the reduced pressure drop over a single contactor, radial-flow contactors of Z-type can be conveniently packed in parallel within a single shell allowing for a straightforward and compact scale-up approach without increasing the adsorption bed thickness and pressure drop.

Radial-flow adsorbers have been studied experimentally and numerically for applications in DAC (Yu and Brilman, 2020; Schellevis et al., 2020), post-combustion $CO_2$ capture (Monteiro et al., 2025), and other air-separation related applications (Xu et al., 2022a,b; Li et al., 2021; Zhang et al., 2022). Yu and Brilman (2020) experimentally demonstrated that a radial-flow adsorber is a promising contactor configuration for DAC because it can combine effective adsorption performance with low pressure drop.

Modelling of DAC is an intricate problem because it involves fluid dynamics coupled with heat and mass transfer across multiple length scales. Being a multiscale process, DAC cannot be adequately captured by a single modelling approach (Marinič and Likozar, 2023): nanoscale methods such as density functional theory describe adsorption mechanisms, binding energies, and force fields; mesoscale approaches such

as Monte Carlo and molecular dynamics use force field parameters to predict statistical and dynamical properties of the system (Xiang et al., 2010); and these insights (often complemented or replaced by experimental data) provide macroscopic kinetic and equilibrium models that are used in fixed-bed dynamic simulations of adsorption. Building on this multiscale foundation, computational fluid dynamics (CFD) becomes crucial for DAC design and optimisation because it resolves the contactor-scale behaviour (flow distribution, pressure drop, temperature fields, and $CO_2$ concentration profiles) that governs the energy demand and productivity of DAC devices (Marinič and Likozar, 2023).

Kinetic and equilibrium models are the essence of modelling the adsorption-based DAC since they describe the mass transfer process between the solid adsorbent and the gas phase (Li et al., 2018). An adsorption kinetic model quantifies how the amount of gas taken up by the sorbent changes with time as a result of mass transfer between the bulk gas and the adsorbent particles, i.e., it defines the adsorption rate. Typical modelling approaches include the local equilibrium model, the pore diffusion models, and the linear driving force (LDF) model (Li et al., 2018). The local equilibrium model is the simplest kinetic model that assumes that the mass transfer occurs instantaneously, ignoring the mass transfer resistance completely; therefore, it has limited applications (Shafeeyan et al., 2014). The pore diffusion models represent the mass-transfer resistance by treating $CO_2$ transport inside the porous sorbent as diffusion through pores, which provides more reliable predictions of the adsorption kinetics. Different mathematical formulations exist for different conditions and pore sizes, which make the models mathematically complicated (Shafeeyan et al., 2014). The linear driving force model represents the mass transfer resistance with a mass transfer coefficient, $k_{LDF}$, without resolving the underlying transport mechanisms. In this approach, the adsorption rate is assumed to be proportional to the difference between the adsorption capacity of the adsorbent at equilibrium, $q_{eq}$, and the actual amount of adsorbed $CO_2$, $q$ (Shafeeyan et al., 2014) as:

$$\frac{\partial q}{\partial t} = k_{LDF}\left(q_{eq} - q\right) \tag{1}$$

An adsorption equilibrium model (isotherm model) describes the equilibrium adsorption capacity, $q_{eq}$, i.e., saturated (equilibrium) uptake of a sorbent as a function of gas conditions (typically temperature and either partial pressure or concentration). These models provide the equilibrium target toward which kinetic models (e.g., LDF) relax during adsorption. In practice, they are usually obtained using adsorption isotherm fitting. A comprehensive review of various adsorption isotherm models is available in Al-Ghouti and Da'ana (2020). The models range from a linear one-parameter (Henry's isotherm) to simple multi-parameter isotherms (most notably, Langmuir-type models with extensions to Langmuir-Freundlich and Toth) and complex multi-parameter and multi-component models.

In addition to the mass transfer model, an appropriate pressure drop model is needed to simulate the viscous and inertial loss of momentum caused by the porous medium. For fixed packed beds, the Darcy model, the Blake-Kozeny model, and the Ergun model are the most common (Li et al., 2018).

Adsorption is an exothermic process, and the temperature changes may affect the adsorption equilibrium and therefore the adsorption rate. Thus, heat transfer should be considered even if isothermal boundary and initial conditions are used. Two main approaches are employed for modelling the heat transfer in porous media, which are equilibrium and non-equilibrium models. The equilibrium model neglects the temperature difference between the adsorbent and the gas phase, and only one energy balance equation is solved for the porous zone. In the non-equilibrium heat transfer model, two energy equations are solved separately for the gas and solid phases, taking into account the heat transfer between the phases. Therefore, two temperature fields are obtained for the porous zone (Li et al., 2018).

CFD modelling has been widely employed in studies of $CO_2$ capture by adsorption. In fixed-bed columns, CFD has been used to improve operating conditions for $CO_2$ adsorption from syngas (Abdullah and Qasim, 2016; Qasim et al., 2014), optimise inlet conditions for $CO_2$ capture from flue gas (Aliyu et al., 2025; Ben-Mansour and Qasem, 2018), identify underutilisation of adsorbent (Chen et al., 2019), and optimise temperature regimes for the desorption step (Huang et al., 2025). It has also been used for comparison of adsorbents for DAC and evaluation of building-integrated DAC systems (Shi et al., 2023). Chen et al. (2023) showed that the W-shaped packed bed can significantly reduce energy consumption compared to a conventional packed bed in an indoor DAC unit. Ben-Mansour et al. (2017) used CFD for geometric optimisation of fixed-bed columns and their configurations.

Radial-flow adsorbers have also been studied by CFD. Sefidan and Vepsäläinen (2025) showed that a radial-flow configuration improves the adsorption of pure $CO_2$ compared to a fixed-bed column. Li et al. (2021) performed geometrical optimisation of a centripetal Z-type radial-flow adsorber using CFD to simulate airflow only, without modelling the adsorption process, and then comparing the flow uniformity. Zhang et al. (2022) optimised the annular channel geometry of a centripetal $\pi$-type adsorber for oxygen enrichment. Xu et al. (2022b) modelled binary adsorption of $CO_2$ and $H_2O$ and compared four types of radial-flow gas separation units; for the selected operating conditions and adsorber dimensions, the centrifugal $\pi$-type achieved the highest flow uniformity and the longest $CO_2$ breakthrough time, indicating better utilisation of adsorbent. Across these studies, the LDF model for adsorption kinetics and the Ergun equation for pressure drop are the most popular modelling choices. For adsorption equilibrium, several models have been employed, most often Langmuir-type isotherms. Both equilibrium and non-equilibrium formulations have been used for heat transfer.

Although CFD simulations have been widely used in various studies involving $CO_2$ adsorption, DAC conditions have received limited attention in these studies. Radial-flow adsorbers for DAC have only been studied experimentally. In this work, CFD modelling is used to simulate the adsorption stage of the DAC process based on a proprietary DAC adsorbent in different configurations of a centripetal Z-type radial-flow contactor. Properties of the adsorbent are determined experimentally. The simulation results are used to compare adsorber configurations according to several design criteria, including the scalability of the adsorber. The organisation of this paper is as follows. In Section 2, the methods are explained through four subsections elaborating the characterisation of adsorbent particles followed by the description of the domain and the governing equations for flow, heat and mass transfer, parametric and mesh sensitivity studies. In Section 3, the results are presented starting with the properties of the adsorbent. Then the results of CFD simulations are presented, which are then followed by the results of the parametric study. After the presentation of the results, the findings are discussed in Section 4. Finally, some concluding remarks are made in Section 5.

## 2. Methods

### 2.1. Material characterisation

The DAC adsorbent used in this work is a commercial amine-functionalised resin. Characterisation of the adsorbent is performed to determine its properties and apply them in simulations. Scanning electron microscopy (SEM) is used to determine the adsorbent surface morphology and particle size. The adsorbent surface area and pore structure were determined from BET analysis, as described in Appendix A.

Breakthrough curve analysis is performed to determine the $CO_2$ adsorption capacity of the adsorbent under humid conditions. The mixSorb analyser (3P Instruments mixSorb SHP, Germany) is used to set the flow of an $N_2/CO_2/H_2O$ mixture and send it through the fixed-bed adsorption column, where 0.25 g of dry pretreated adsorbent is loaded. The FTIR gas analyser (Matrix MG5C, Bruker, UK) is used to measure the concentration of $CO_2$ and $H_2O$ after the column. The tests are performed at 10, 25, 40 and 60 °C and at partial pressure of $CO_2$ ranging from 0.15 to 5.5 mbar. The concentration of water vapour in the mixture is kept at

0.8-2 vol-%. The adsorbent is pretreated overnight at 60 °C with dry $N_2$ flow before the start of the tests at each temperature. The adsorption isotherms are obtained from the breakthrough curve analysis after calculating the equilibrium adsorption capacity $q_{eq}$, $mmol_{CO_2}/g_{ads}$ in each test from the $CO_2$ mass balance in the adsorption column:

$$\frac{\partial m_{CO_2}}{\partial t} = \dot{n}_{tot} M_{CO_2} \left(y_{in} - y_{out}(t)\right) \tag{2}$$

where $\dot{n}_{tot}$ is the total molar flow rate, $y_{in}$ and $y_{out}$ are the molar fraction of $CO_2$ at the column inlet and outlet. Rewriting Eq. (2) for the amount of $CO_2$ and integrating it gives the expression for the amount of adsorbed $n_{CO_2}$:

$$n_{CO_2} = \dot{n}_{tot} \int_{t_0}^{t_1} \left(y_{in} - y_{out}(t)\right) dt \tag{3}$$

Then the equilibrium adsorption capacity is calculated as

$$q_{eq} = \frac{n_{CO_2}}{m_{ads}} \tag{4}$$

where $m_{ads}$ is the mass of the adsorbent in the column.

The breakthrough curve analysis is also used to approximate the mass transfer rate coefficient $k_{LDF}$ in the linear driving force model (Eq. (1)). Using the definition of the adsorption capacity (Eq. (4)), the rate of $CO_2$ adsorption can be presented as

$$\frac{\partial m_{CO_2}}{\partial t} = m_{ads} M_{CO_2} \frac{\partial q}{\partial t} \tag{5}$$

Substituting the LDF model (Eq. (1)) gives

$$\frac{\partial m_{CO_2}}{\partial t} = k_{LDF} m_{ads} M_{CO_2} \left(q_{eq} - q(t)\right) \tag{6}$$

From Eqs. (2) and (6), an expression is derived for calculating the mass transfer rate coefficient at a given time $t$:

$$k_{LDF} = \frac{\dot{n}_{tot}\left(y_{in} - y_{out}(t)\right)}{m_{ads}\left(q_{eq} - q(t)\right)} \tag{7}$$

The density of adsorbent particles is measured at room temperature using the water displacement method. The C-Therm Trident thermal conductivity analyser is used to measure the thermal conductivity $k$ and thermal effusivity $e$ of the adsorbent. The specific heat capacity is then calculated as

$$c_p = \frac{e^2}{\rho k} \tag{8}$$

### 2.2. Computational fluid dynamics model

The 2D axisymmetric CFD simulations of $CO_2$ adsorption in a single radial-flow contactor are performed using Ansys Fluent 2022 R2. The computational domain consists of three parts (Fig. 1): an inflow channel, an adsorption bed, and the outflow channel at the axis of the device, so air moves through the bed in the radial direction from the outer to the inner side of the zone.

**Table 1**
Governing equations.

| | |
|---|---|
| Continuity equation | |
| $\nabla \cdot (\rho \vec{v}) = 0$ | (9) |
| Momentum equation | |
| $\nabla \cdot (\rho \vec{v}\vec{v}) = -\nabla p + \nabla \cdot \left[\mu\left(\nabla \vec{v} + \nabla \vec{v}^{\mathrm{T}}\right)\right] - \left(\frac{\mu}{\alpha}\vec{v} + C_2 \frac{1}{2}\rho \lvert\vec{v}\rvert \vec{v}\right)$ | (10) |
| $CO_2$ transport equation | |
| $\frac{\partial}{\partial t}\left(\varepsilon \rho Y_{CO_2}\right) + \nabla \cdot \left(\rho \vec{v} Y_{CO_2}\right) = -\nabla \cdot \vec{J}_{CO_2} + S_{CO_2}$ | (11) |
| Energy equation for the fluid zone | |
| $\frac{\partial}{\partial t}\left(\varepsilon \rho_f E_f\right) + \nabla \cdot \left(\vec{v}\left(\rho_f E_f + p\right)\right) = \nabla \cdot \left(\varepsilon k_f \nabla T_f - h_{CO_2} \vec{J}_{CO_2}\right) + h_{fs} A_{fs}\left(T_s - T_f\right) + S_E$ | (12) |
| Energy equation for the solid zone | |
| $\frac{\partial}{\partial t}\left((1-\varepsilon)\rho_s E_s\right) = \nabla \cdot \left((1-\varepsilon) k_s \nabla T_s\right) + h_{fs} A_{fs}\left(T_f - T_s\right)$ | (13) |
| Diffusion flux | |
| $\vec{J}_{CO_2} = \rho D_{m,CO_2} \nabla Y_{CO_2}$ | (14) |

The air is treated as a two-component ideal gas mixture of $N_2$ and $CO_2$. The flow is assumed laminar. The concentration of $CO_2$ in the mixture is obtained by solving the species transport equation. The adsorption bed zone is represented by a model for a porous medium. Non-equilibrium thermal model for porous medium is implemented using the dual-cell approach, in which a new solid zone is defined spatially coinciding with the porous-fluid zone. This solid zone is coupled to the fluid through interphase heat transfer. Separate energy conservation equations are then solved for the fluid and solid regions to allow separate gas- and solid-phase temperature fields to be obtained during the exothermic adsorption process.

The equations used to model the adsorption bed are presented in Table 1. Due to a low concentration of $CO_2$ in the mixture and small temperature variations during adsorption, the evolving $CO_2$ concentration and temperature fields have a negligible effect on the flow throughout the device. Thus, a one-way coupling from the flow field to the species transport and energy equations is applied: the steady-state continuity (Eq. (9)) and momentum (Eq. (10)) equations are first solved to obtain the flow field. Then, the transient $CO_2$ transport (Eq. (11)) and energy equations (Eqs. (12) and (13)) are solved using the obtained flow field.

The adsorption of $CO_2$ is modelled using the source terms in the $CO_2$ transport and energy equations (Eqs. (11), (12)). The $CO_2$ mass sink is defined as

$$S_{CO_2} = -\frac{m_{ads} M_{CO_2}}{V_{cell}} \frac{\partial q}{\partial t} \tag{15}$$

and the source term in the energy equation accounts for the $CO_2$ enthalpy sink and adsorption heat source based on the amount of adsorbed $CO_2$:

$$S_E = S_{CO_2}\left(c_p\left(T - T_{ref}\right) - \frac{\Delta H}{M_{CO_2}}\right) \tag{16}$$

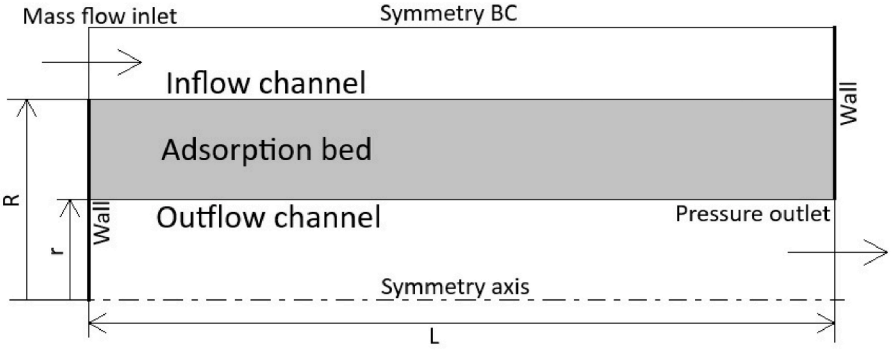



**Fig. 1.** Computational domain.

The $CO_2$ adsorption rate in Eq. (15) is calculated using the linear driving force model (Eq. (1)), which is selected as a well-established approach for representing adsorption kinetics for contactor-scale CFD simulations without resolving particle-scale mass transport. The equilibrium adsorption capacity is calculated using the Toth model (Keller and Staudt, 2005) because it provided a good fit to the experimentally obtained adsorption isotherms over the investigated conditions:

$$q_{eq} = \frac{q_m b p_{CO_2}}{\left[1 + \left(b p_{CO_2}\right)^{t_{Toth}}\right]^{\frac{1}{t_{Toth}}}} \tag{17}$$

The viscous resistance $1/\alpha$ and inertial resistance $C_2$ in Eq. (10) are calculated using the Ergun equation, a well-established and widely used correlation for momentum resistance and pressure loss in randomly packed beds (Ergun, 1952):

$$1/\alpha = 150\frac{(1-\varepsilon)^2}{d_p^2 \varepsilon^3} \tag{18}$$

$$C_2 = 3.5\frac{(1-\varepsilon)}{d_p \varepsilon^2} \tag{19}$$

The fluid-solid heat transfer coefficient in Eqs. (12) and (13) is calculated using the Wakao and Kaguei correlation (Wakao et al., 1979):

$$\mathrm{Nu} = 2 + 1.1\mathrm{Re_p}^{0.6}\mathrm{Pr}^{1/3} \tag{20}$$

where

$$\mathrm{Nu} = \frac{h_{fs} d_p}{k_f} \tag{21}$$

$$\mathrm{Re_p} = \frac{\rho d_p v_\infty}{\mu} \tag{22}$$

The interfacial area density $A_{fs}$ is calculated as

$$A_{fs} = \frac{6(1-\varepsilon)}{d_p} \tag{23}$$

The mass diffusivity $D_{m,CO_2}$ in Eq. (14) is calculated using kinetic theory (a modification of the Chapman-Enskog formula Bird et al., 2002).

The boundary conditions are shown in Fig. 1. The mass flow inlet is set with a constant mass flow rate, temperature and concentration of $CO_2$. The pressure outlet is set for the outlet. All walls are adiabatic and subject to no-slip boundary conditions. The symmetry boundary condition at the outer side of the inflow channel implies that there can be more contactors placed in parallel to the DAC unit.

The inlet temperature and the initial temperature in the whole domain are set to 20 °C. The inlet concentration of $CO_2$ is 430 ppm. The initial concentration of $CO_2$ is zero in the domain and no $CO_2$ is adsorbed in the material. The inlet mass flow rate is discussed in the next subsection.

The simulations are performed using the finite volume method implemented in Ansys Fluent's pressure-based solver. For the steady-state flow simulations, the pressure-velocity coupling is done using the coupled algorithm. For the transient solution of the $CO_2$ transport and energy equations, the first-order implicit transient formulation is employed. For the spatial discretisation of density, momentum, $CO_2$ concentration, and energy, a second-order upwind scheme is utilised. The second-order scheme is implemented for pressure interpolation. The least squares cell-based method is applied for the gradient evaluation.

The steady-state simulations are performed using Eqs. 9 and (10) until the solution is converged. Then the transient simulations are performed using Eqs. (11) to (14) with a time step of 0.1 s and 20 iterations per time step during the first 100 s and 6 iterations per time step after that.

### 2.3. Parametric study

The CFD model described above is used for the parametric study of the DAC unit. The goal of the study is to compare different geometrical configurations of the contactor in terms of productivity, energy requirements of the adsorption step, the size of the adsorber, and its scalability.

The base case is selected with the following dimensions (Fig. 1): $L = 0.48$ m, $R = 0.12$ m, $r = 0.06$ m. Two geometric parameters are varied to set up the other cases for the analysis, namely the bed aspect ratio and the radius ratio,

$$\mathrm{AR} = \frac{L}{R-r} \tag{24}$$

$$\mathrm{RR} = \frac{r}{R} \tag{25}$$

For the base case, AR = 8 and RR = 0.5. The bed aspect ratio is varied between 4, 8, 16, and 32, and the radius ratio between 0.3, 0.5, and 0.7. The volume of the bed and the mass of the adsorbent remain the same as in the base case. The dimensions of the domain are calculated for each case based on this condition. Illustrations of all 12 cases are presented in Appendix B.

The mass flow rate of the gas mixture of 0.02423 kg/s is chosen to obtain a pressure drop of 200 Pa for the base case. The width of the inlet channel for the base case is 0.03 m. For the other cases, it is changed to keep the inlet velocity the same as in the base case.

### 2.4. Mesh sensitivity study

The mesh sensitivity study is performed to ensure that the solution is grid independent. The mesh sensitivity analysis focused on the base case. First, a structured mesh containing 18,000 cells (excluding the additional solid porous zone) is created, and the transient simulations are performed with different time steps. The results show that as the time step falls below 0.1 s, the solution does not display distinguishable differences from that corresponding to 0.1 s. However, if the time step is larger than 0.2 s, the convergence of simulations fails. Based on these results, the time step size of 0.1 s is found to work for all transient simulations.

For the mesh sensitivity study, four more meshes are created with 9,010, 12,726, 25,650, and 35,595 grid cells. In each mesh, steady-state simulations of the flow are first performed to compare the pressure drop, which appears to be the same ($200 \pm 0.02$ Pa) in all cases. Then, transient simulations of adsorption are performed for which the time to reach 95% saturation of the bed is obtained. Compared to the finest mesh, the other meshes overestimated the time by 0.7%, 1.1%, 1.8%, and 3.1% (in descending order of the number of cells). Based on these results, the original 18,000 grid cells are selected for the base case as providing acceptable accuracy. For the other cases in the parametric study, the meshes are created with the same mesh size as in the selected mesh.

## 3. Results

The results of the current study as well as the methods of analysis of the results are presented in this section and discussed in the next section.

### 3.1. Material characterisation

A representative particle diameter of 0.591 mm is obtained from the SEM analysis. An SEM image of the adsorbent particles is presented in Fig. 2 with a magnification factor of 50. It can be seen that the particles have a spherical shape and a macroporous surface. The BET surface area and pore properties are presented in Appendix A.

The adsorption isotherms are experimentally obtained (symbols) and fitted (solid lines) to the Toth model (Eq. (17)) at four different temperatures, as shown in Fig. 3. The corresponding coefficients of determination ($R^2$) range from 0.9978 to 0.9989 for the investigated isotherms.

Other measured and calculated properties of the adsorbent and the adsorption bed that are used in the simulations are presented in Table 2.

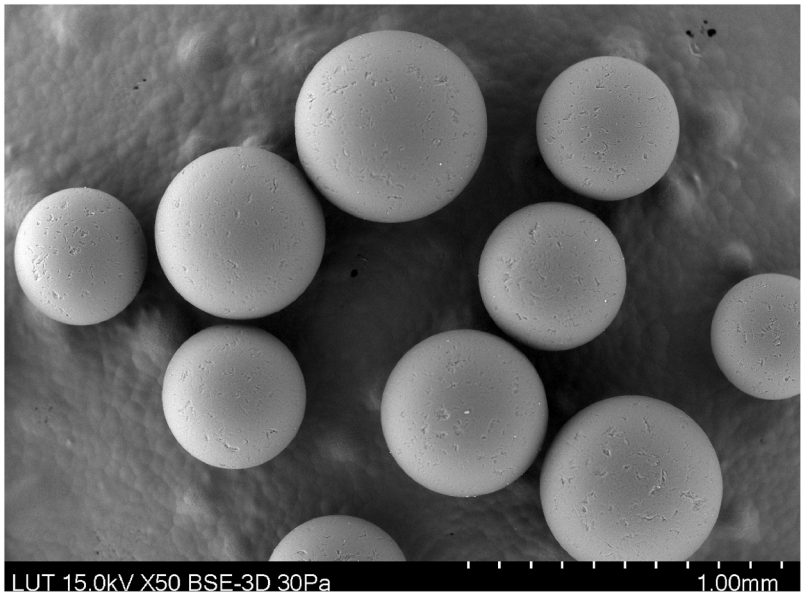



**Fig. 2.** SEM-image of the adsorbent particles.

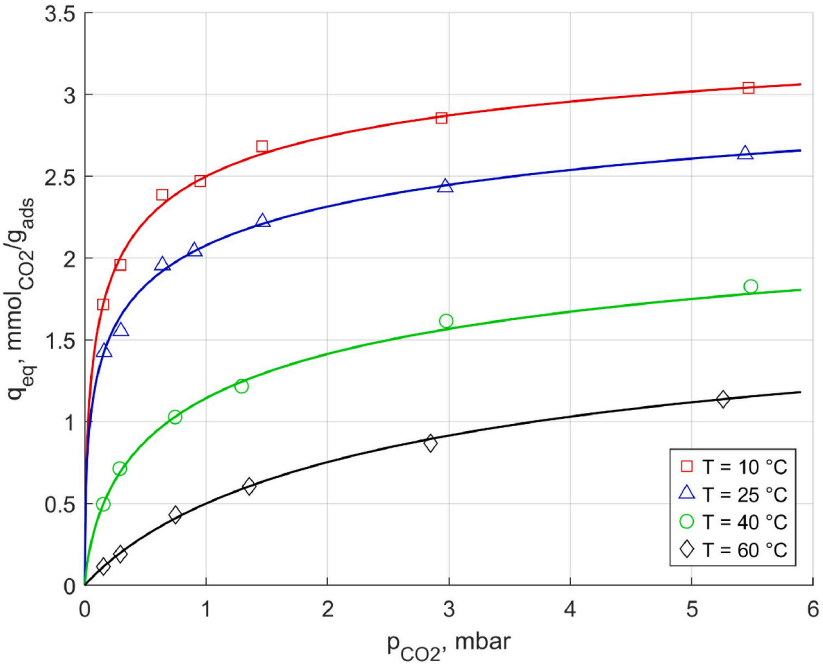



**Fig. 3.** Adsorption isotherms fitted to the Toth model. Symbols represent the experimental data and solid lines of the same colours as the symbols represent the fits.

### 3.2. CFD simulations

The breakthrough curves obtained from the transient CFD simulations, i.e., the normalised $CO_2$ concentration at the outlet, are presented for all cases in Fig. 4. Typically, the breakthrough starts after 10 h of simulation and the full saturation is achieved by 14 h. However, several cases do not follow the typical curves. For instance, the case corresponding to AR = 32 and RR = 0.3 does not approach saturation within the simulated time of 20 h. To illustrate the differences between the typical cases and the deviants, the spatial distribution of pressure, radial velocity, and $CO_2$ concentration in the adsorption bed is shown during the adsorption process for the base case and the most deviating case in Figs. 5 and 6. The $CO_2$ concentration is shown through multiple snapshots at different instants.

### 3.3. Parametric study

The main results of CFD simulations used for comparison of the investigated cases are presented in Fig. 7 in the form of 12 heatmaps. Some of them represents the main geometric parameters calculated from Eqs. (24) and (25) including the length, the inner radius, and the outer radius of the bed (Fig. 7a–c). The thickness of the bed (Fig. 7d) is also included, as it has a significant effect on the pressure drop. In addition, the total surface area of the contactor is calculated (Fig. 7e). The surface

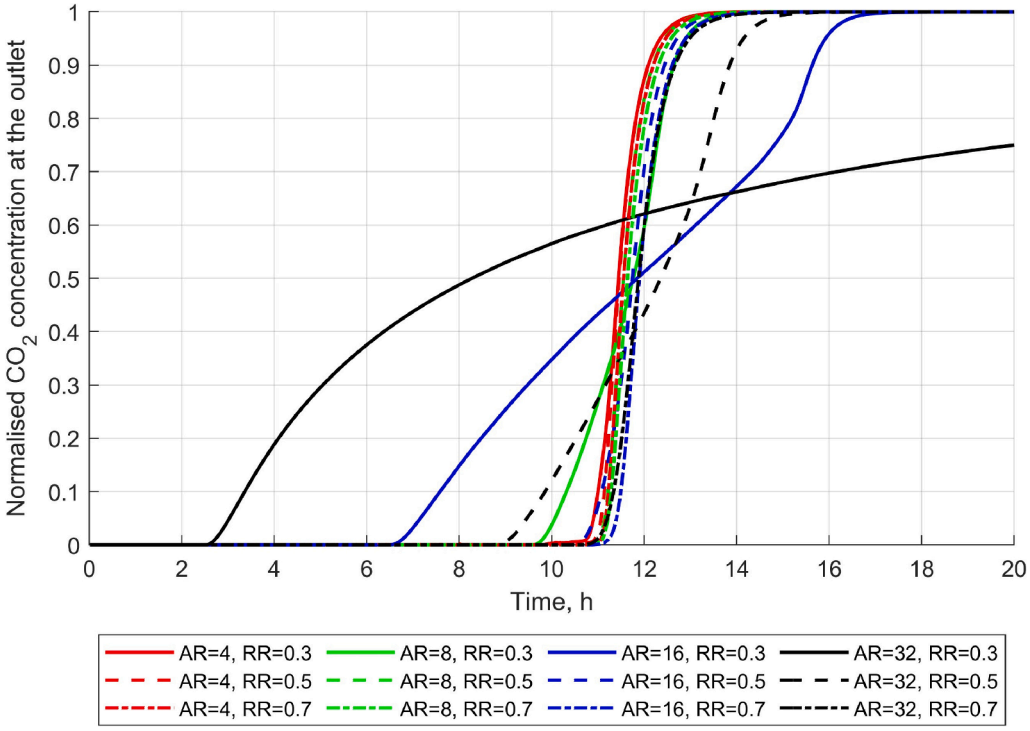



**Fig. 4.** Temporal variation of $CO_2$ concentration at the outlet in all cases.

**Table 2**
Material properties.

| Property | | Units | Value |
|---|---|---|---|
| Mass transfer coefficient | $k_{LDF}$ | $s^{-1}$ | 0.001 |
| Particle diameter | $d_p$ | m | $5.91 \cdot 10^{-4}$ |
| Porosity | $\epsilon$ | | 0.4 |
| Density | $\rho$ | kg $m^{-3}$ | 1272 |
| Thermal conductivity | $k$ | W $m^{-1}K^{-1}$ | 0.146 |
| Specific heat capacity | $c_p$ | J $kg^{-1}K^{-1}$ | 1009.6 |
| Viscous resistance | $1/\alpha$ | $m^{-2}$ | $2.42 \cdot 10^{9}$ |
| Inertial resistance | $C_2$ | $m^{-1}$ | 22,208 |
| Fluid-solid heat transfer coefficient | $h_{fs}$ | $Wm^{-2}K^{-1}$ | 168.4 |
| Interfacial area density | $A_{fs}$ | $m^{-1}$ | 6091.4 |

area can serve as a geometric indicator of construction material demand in comparative studies where the volume of the bed (and the amount of adsorbent) is kept constant; wall thickness and detailed structural design are not considered in this metric.

Meanwhile, the results of the steady-state flow simulations are used to calculate the pressure drop across a single contactor as the difference between the inlet and outlet pressure (Fig. 7f). To quantitatively assess the differences in pressure distribution, which lead to non-uniform bed saturation during adsorption, a flow uniformity index is calculated (Fig. 7g) adopting from Xu et al. (2022b) given by:

$$\eta = \sqrt{\frac{\Delta p_{in}}{\Delta p_{out}}} \tag{26}$$

where $\Delta p_{in}$ and $\Delta p_{out}$ are the pressure drops over the bed in the radial direction near the inlet and outlet, respectively.

The adsorption time (Fig. 7h) is defined as the time to reach 95% saturation of the adsorption bed. Since the case of AR = 32 and RR = 0.3 did not reach the 95% saturation threshold, it is excluded from further analysis.

The specific energy requirement (SER, $MJ/kg_{CO2}$) represents the energy needed for the adsorption of 1 kg of $CO_2$ (Fig. 7i). It is calculated using the following equation:

$$\mathrm{SER} = \frac{\Delta p \dot{m} t_{sat}}{\rho m_{CO_2}} \tag{27}$$

where $\Delta p$ is the pressure drop, $\dot{m}$ and $\rho$ are the mass flow rate and the density of the gas mixture, $t_{sat}$ is the adsorption time, $m_{CO_2}$ is the mass of $CO_2$ adsorbed during the time $t_{sat}$.

The productivity of the adsorber (Fig. 7j) is calculated using the adsorption time. It is defined as the mass of $CO_2$ adsorbed per hour per kg of adsorbent.

The last set of results (Fig. 7k-l) considers the possibility of scaling up the DAC device by placing several contactors in parallel within a common shell as shown in Fig. 8. Here, a 7-contactor hexagonal configuration is selected, with one contactor in the middle and six others surrounding it, as it provides the most compact array of contactors desired in the design and manufacturing of the shell. The total volume of such an adsorber is calculated by assuming an annular gap of 0.02 m around each contactor in all cases. This volume is an important parameter as it affects the space requirements; however, to assess the practicality of different scaled-up configurations, the normalised aspect ratio (smaller-to-larger dimension ratio) is calculated for the 7-contactor adsorber. Very low values of the normalised aspect ratio indicate that either the adsorber diameter or the length is too big and can be con-

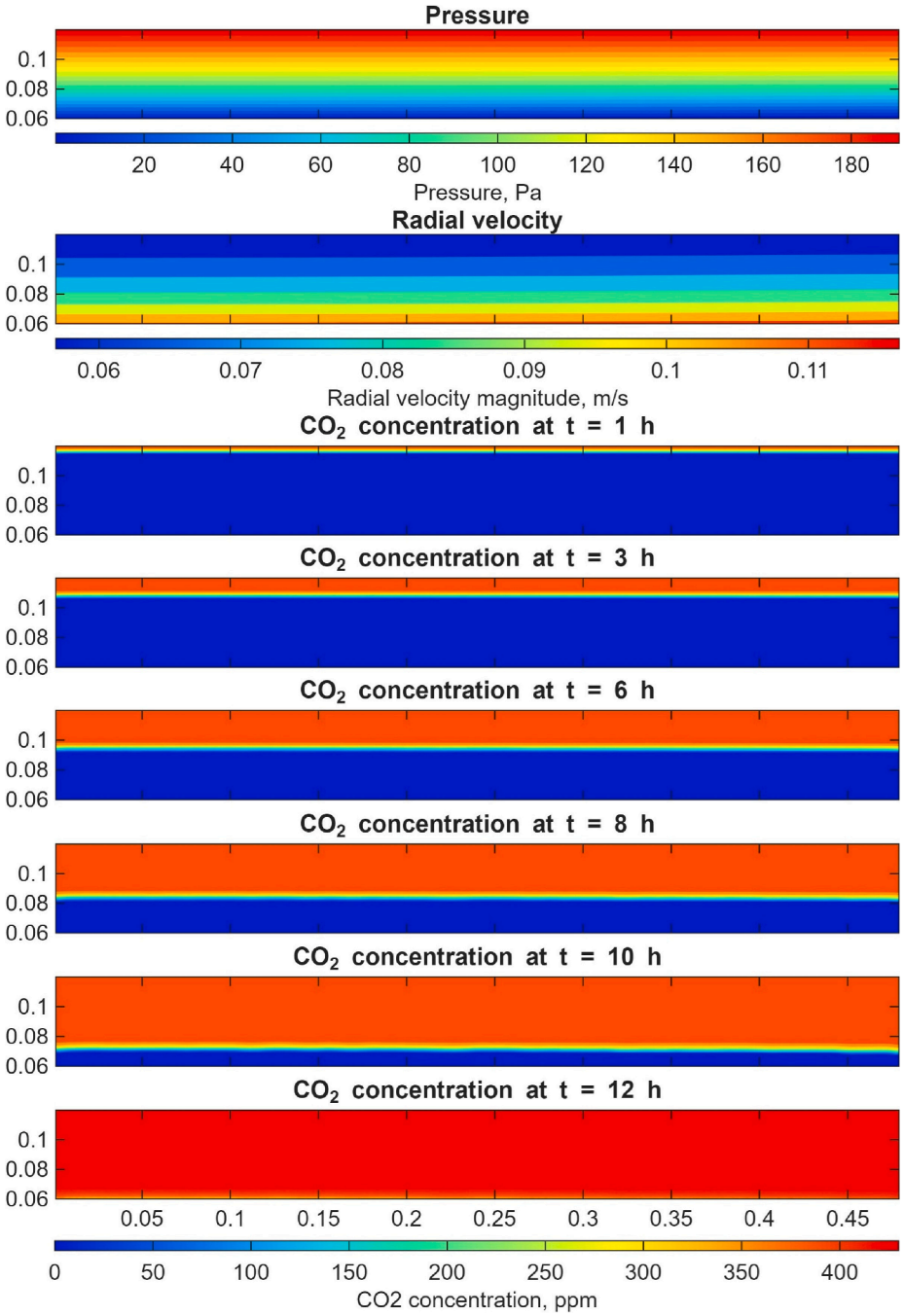



**Fig. 5.** Spatial distribution of pressure, radial velocity, and $CO_2$ concentration in the adsorption bed in the base case (AR = 8, RR = 0.5), not to scale.

sidered impractical. It should be mentioned that any contactor is assumed not to disturb the flow, heat, and mass transfer in its surrounding contactors.

## 4. Discussion

For simplicity, the configurations studied in this work are referred to in the text in the form AR_RR. For example, case 8_0.5 refers to a configuration with a bed aspect ratio of 8 and a radius ratio of 0.5.

The deviation of the breakthrough curves from the common pattern seen in Fig. 4 is caused by the non-uniform flow distribution. It can be noted that for all deviant cases (32_0.3, 16_0.3, 32_05, 8_0.3), the greater the deviation, the lower the flow uniformity coefficient (Fig. 7g). Reduced flow uniformity means that the pressure drop over the bed on the inlet side of the contactor is lower than that on the outlet side (Eq. (26)). This is clearly seen in case 32_0.3 (Fig. 6) compared to case 8_0.5 (Fig. 5) in which the pressure distribution is uniform. In the inflow channel, the pressure does not change significantly along the bed, so this pressure drop variation is caused by the high pressure gradient in the outflow channel. This can be explained by the dimensions of the inner (axial) pipe (Fig. 7a-b). The configurations with low flow uniformity have a smaller inner pipe radius and a larger length, causing high flow resistance and consequently high pressure drop over the inner pipe. These findings are consistent with Li et al. (2021), who showed that the dimensions of the central and annular flow channels strongly affect flow uniformity in a Z-type radial-flow adsorber. However, while their analysis focused on pressure uniformity, the present results show that configurations with the highest flow uniformity are not necessarily preferred

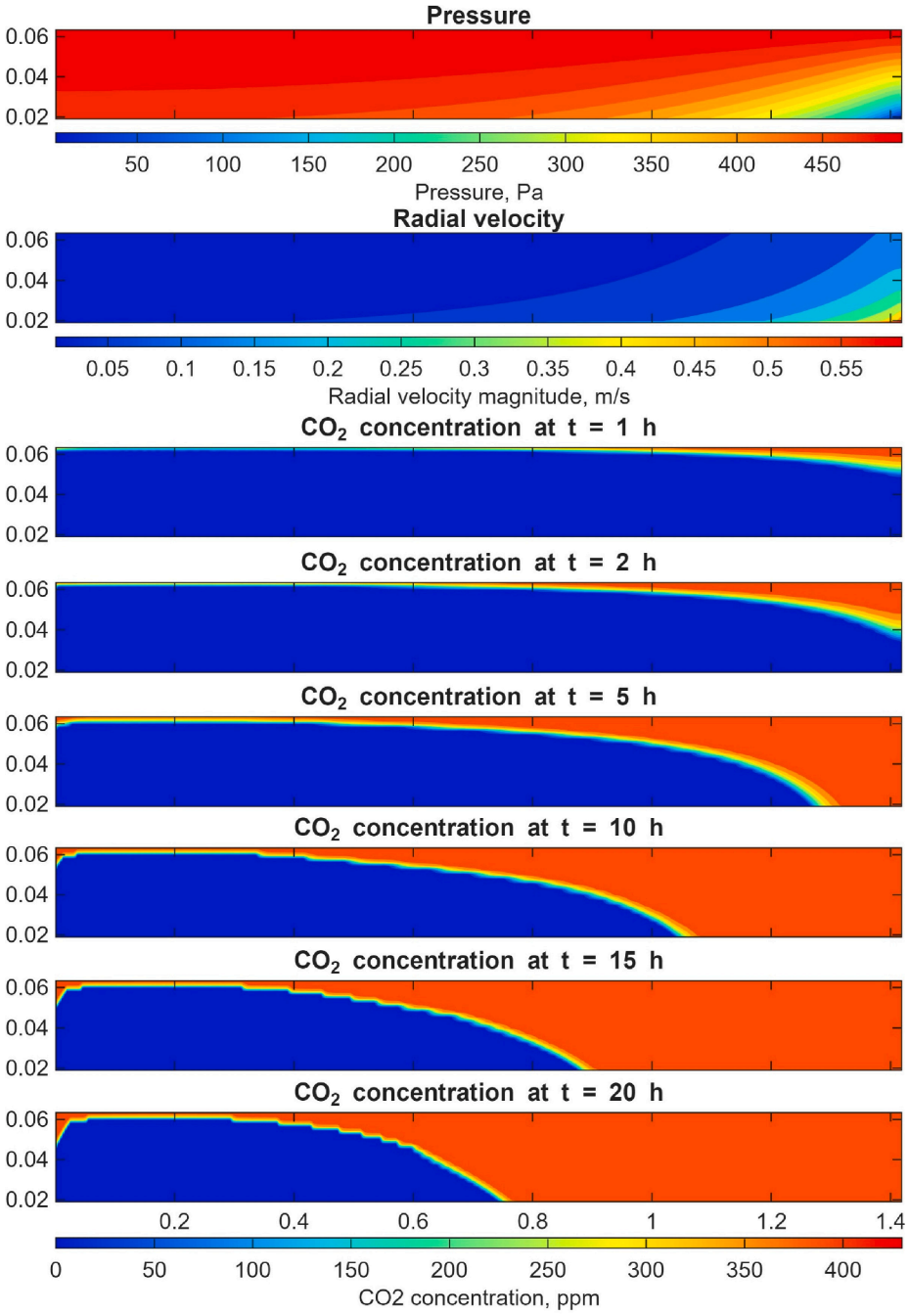



**Fig. 6.** Spatial distribution of pressure, radial velocity, and $CO_2$ concentration contours in the adsorption bed in the case of AR = 32, RR = 0.3, not to scale.

when adsorption performance, specific energy requirements, and scale-up constraints are considered simultaneously.

The velocity of flow through porous media increases with pressure drop; therefore, if the pressure drop is higher near the outlet, the velocity of the gas mixture and its flow rate are also higher, as seen in Fig. 6. This is why adsorption occurs at different rates on the inlet and outlet sides of the contactor in the deviant cases, causing a non-uniform distribution of $CO_2$ concentration (Fig. 6). As a result, part of the adsorbent on the inlet side takes a longer time to reach the saturation. The breakthrough curve becomes less steep, and the total adsorption time increases (Fig. 7h). Otherwise, the variation of the adsorption time would be small as it mostly depends on the mass flow rate, which is kept constant in this work.

With the selected definition of productivity [$g_{CO2}/(kg_{ads}h)$], it depends only on the adsorption time in this study, because the mass of the adsorbent and the adsorbed $CO_2$ are the same in all cases. Therefore, increased adsorption time due to non-uniform flow leads to reduced productivity (Fig. 7j). Cases with low productivity can still be considered for DAC if the adsorption stage is interrupted when the adsorption rate starts to decrease. This approach could keep the productivity high, but it leads to underutilisation of the adsorbent, which should be avoided.

A high pressure drop strongly correlates with the configurations that have large bed thickness, except for the deviant cases of low flow uniformity. In these cases, the pressure drop is higher than expected on the basis of the bed thickness alone. The pressure drop is increased by an additional pressure drop over the axial outflow channel.

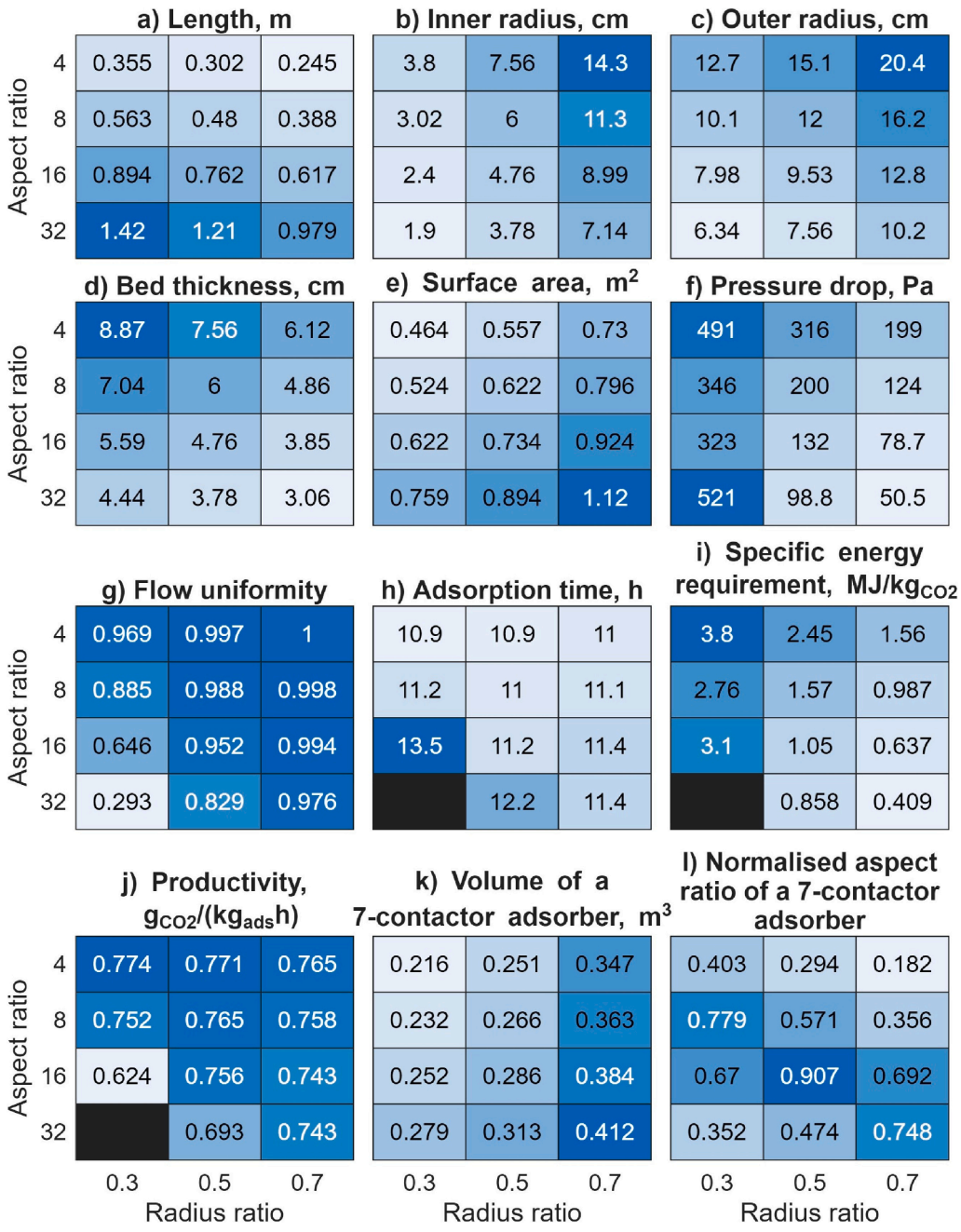



**Fig. 7.** Geometric parameters, flow and adsorption performance indicators for all simulated cases.

The results show that the specific energy requirement is strongly dependent on the pressure drop with small additional increases in cases with a high adsorption time. The case with the smallest bed thickness (32_0.7) has the lowest pressure drop and SER, while the cases with low AR and RR have higher SER. It is clear that if SER is the only criterion for the selection of contactor geometry, preference should be given to configurations with the lowest bed thickness and high flow uniformity.

On the other hand, the surface area is the smallest in the case of low AR and RR (4_0.3), and increases as AR and RR rise. That must be taken into account, especially in situations where the relative importance of SER as a criterion for configuration selection is low, e.g., in regions with low-cost renewable electricity and high construction material costs. A larger surface area would require higher costs of construction material.

The volume of a 7-contactor adsorber (Fig. 7k) follows the same pattern as the surface area: increasing AR and RR leads to a larger volume. This means that the most energy-efficient configuration (32_0.7) will require the largest space, and vice versa, case 4_0.3, which has the highest SER, requires almost half as much space. However, as is noticeable from the low values of the normalised aspect ratio of a 7-contactor adsorber (Fig. 7l), the cases with low AR are quite impractical. Because of the large outer radius of the bed, the scaled-up adsorber with 7 parallel contactors has a very large shell diameter at a relatively small length. Configurations with high AR and low RR also have a small normalised aspect ratio of a scaled-up adsorber because the length is much larger than the total diameter. A very long or wide configuration is generally undesirable because it is often harder and more expensive to fabricate, transport, install, and maintain, especially at a larger scale. Inclusion of

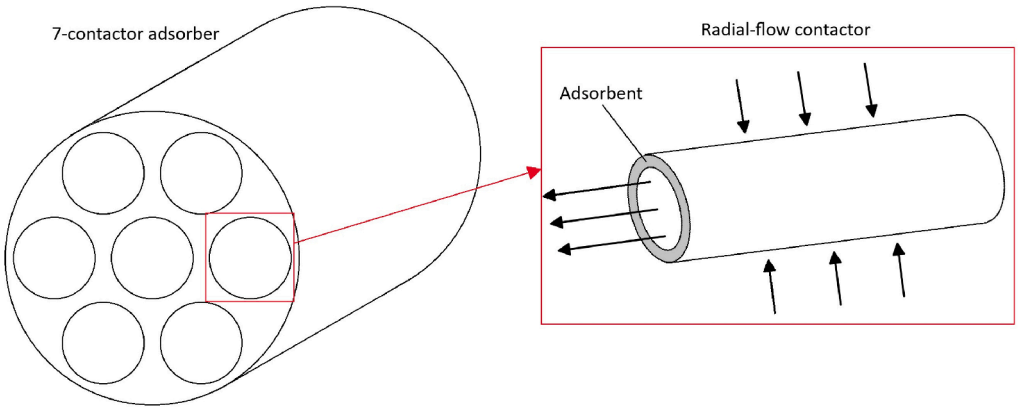



**Fig. 8.** Illustration of a scaled-up 7-contactor adsorber.

these cases in this work is simply to demonstrate how poorly they could perform as a DAC unit.

To better illustrate the trade-off between the specific energy requirement and geometrical constraints, Fig. 9 presents all the cases on the SER–surface area plane (Fig. 9a) and SER–volume plane (Fig. 9b). The marker colour shows the normalised aspect ratio of a 7-contactor adsorber for each case to take into account the practicality of scaling up in each case. These graphs are crucial because they show how much an increase in SER is required to achieve a lower surface area and volume of a scaled-up adsorber. For example, if AR is reduced from 32 to 16 at RR = 0.7, the surface area decreases by 0.2 $m^2$ at the cost of increasing the SER by 0.2 $MJ/kg_{CO2}$. The volume of a 7-contactor adsorber is then slightly reduced from 0.41 to 0.38 $m^3$. The trade-off between material costs and energy requirements should be interpreted further for specific conditions of a DAC plant in terms of the levelised cost of $CO_2$ capture that must be minimised.

It is important to note that Fig. 9 uses the SER calculated for a single contactor for the comparison of configurations. For a 7-contactor adsorber, it is assumed that the 7-times higher mass flow rate is evenly divided between the contactors, although unequal flow splitting between the contactors may lead to differences in their adsorption behaviour. An additional pressure drop caused by the separation and expansion of the flow can slightly increase the adsorption stage SER. However, the thickness of the adsorption bed remains the main contributor to pressure drop, and the parallel placement of radial-flow contactors allows to keep the thickness fixed. To assess the flow maldistribution between the contactors and determine the additional pressure drop, 3D simulations of the flow through a 7-contactor adsorber are needed; however, it is not in the scope of this paper. The conclusions on the practicality of the scale-up are made only on the basis of geometric considerations.

Several modelling assumptions and limitations of the present study should be taken into account. The adsorption bed is assumed to be stationary, homogeneous, and isotropic and is represented as a porous medium with a constant effective porosity. Consequently, spatial variations in porosity arising from the local particle packing are not resolved. Individual adsorbent particles and flow between them are also not explicitly resolved; instead, particle-scale mass, momentum, and heat-transfer are represented in an averaged manner through the LDF model and relevant packed-bed closure correlations. This simplification is consistent with the comparative scope of the present study, in which the same bulk bed properties are applied to all investigated geometries. Water adsorption is also not explicitly included in the CFD model. However, the $CO_2$ equilibrium and kinetic parameters were derived from breakthrough experiments performed under humid conditions and therefore provide an effective representation of the influence of humidity on $CO_2$ adsorption under the investigated conditions. Competitive $CO_2/H_2O$ adsorption and the associated heat effects are not resolved and would require additional multicomponent equilibrium and kinetic data. Another modelling assumption is that the flow regime is considered laminar throughout the domain. Although turbulent conditions were reached in the outflow channel for some configurations, additional simulations using the SST k-ω turbulence model showed only minor effects on the results in most cases. A larger difference occurred only for configuration 32_03, where the laminar model overestimated the pressure drop by 8.6%; however, the turbulent simulation reached only 69% bed saturation after 20 h compared to 64% in the laminar simulation. Thus, this configuration would still be excluded from the comparative analysis. Finally, a limitation of the present study is that the CFD model has not been validated against breakthrough data from a radial-flow contactor. Therefore, the absolute values of the simulated adsorption performance should not be interpreted as experimentally validated predictions. The model is used primarily here for a comparative assessment of the investigated geometries. Since the same modelling assumptions and operating conditions are applied consistently in all cases, the results provide useful insight into the relative trends in flow distribution, pressure drop, adsorption time, specific energy requirement, and productivity associated with variations in contactor geometry.

The comparison of the cases can be summarised as follows. Cases with both low AR and RR possess the smallest surface area and volume of a 7-contactor adsorber, but the highest SER. Cases with high AR and RR have the smallest SER but the largest surface area and volume. Both extremes have decent flow uniformity and, therefore, adequate adsorption time and productivity. Cases with low AR and high RR allow for the most uniform flow; however, other configurations achieve equal or lower values of SER, surface area, and volume simultaneously. In addition, these cases have the lowest normalised aspect ratio of a 7-contactor adsorber, worsening the scalability prospects. Cases with high AR and low RR should not be considered because of their significant flow non-uniformity. Configurations in the middle of the case matrix, e.g. 16_0.5, represent a compromise between SER and geometric parameters and achieve good flow uniformity and the best normalised aspect ratio of a 7-contactor adsorber.

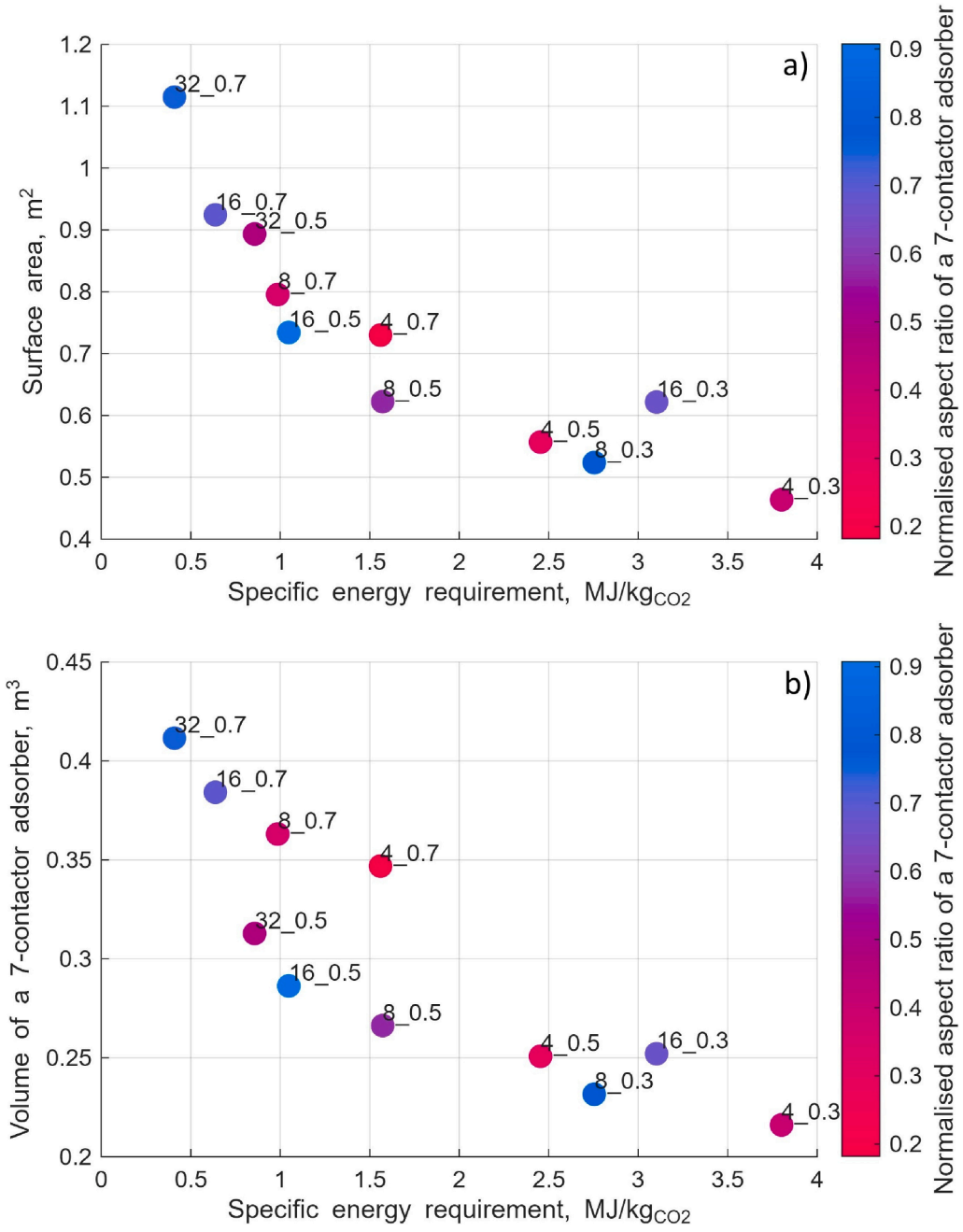



**Fig. 9.** Trade-off between SER and a) contactor surface area, b) volume of a 7-contactor adsorber (case labels show the bed aspect ratio and the radius ratio as AR_RR).

## 5. Conclusions

Direct air capture is expected to play an important role in carbon dioxide removal, but its large energy demand and cost remain major barriers to deployment. Improving the design of DAC contactors is therefore important for reducing energy consumption of the process and for developing configurations capable of being scaled up in a practical way. Radial-flow adsorbers are advantageous in this respect because they can combine low pressure drop with compact packed-bed design, but their optimisation for DAC has received limited attention.

In this work, a two-dimensional axisymmetric CFD model was developed to simulate the adsorption stage of DAC in a centripetal Z-type radial-flow contactor. The properties of a commercial DAC adsorbent were determined experimentally and used as model inputs. A geometric parametric study was then performed by varying the bed aspect ratio and radius ratio while keeping the bed volume constant.

The results showed that the contactor geometry affects pressure drop, flow uniformity, adsorption time, specific energy requirement, and scale-up characteristics. Configurations with a high bed aspect ratio and a high radius ratio resulted in the lowest specific energy requirement because of the smaller bed thickness, while configurations with a low bed aspect ratio and a low radius ratio yielded the smallest surface area and volume of a scaled-up adsorber but with the highest energy demand. The intermediate configurations provided a compromise between energy performance, geometric constraints, and scale-up practicality. These results provide preliminary design guidance for radial-flow DAC adsorbers and reaffirm that CFD is an effective tool for their multi-criteria assessment.

## CRediT authorship contribution statement

**Konstantin Zaynetdinov:** Writing – original draft, Visualization, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization; **Eero Inkeri:** Writing – review & editing, Supervision, Resources, Investigation, Conceptualization; **Aafaq Rehman:** Writing – review & editing, Supervision, Resources, Investigation; **Payman Jalali:** Writing – review & editing, Validation, Supervision, Conceptualization; **Tero Tynjälä:** Writing – review & editing, Supervision, Project administration, Funding acquisition, Conceptualization.

## Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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## Data availability

Data will be made available on request.

## Appendix A. Adsorbent surface area and pore properties

The specific surface area and pore properties of the adsorbent were determined by $N_2$ adsorption at 77 K using a Micromeritics 3Flex 3500 surface area and porosity analyser with a VacPrep 061 sample preparation unit. The specific surface area was determined using the Brunauer–Emmett–Teller (BET) method. The obtained properties are summarised in Table A.1.

**Table A.1**
Surface area and pore properties of the adsorbent.

| Property | Units | Value |
|---|---|---|
| BET specific surface area | $m^2/g$ | 30.186 |
| Total pore volume | $cm^3/g$ | 0.1127 |
| Average pore diameter | nm | 14.934 |

## Appendix B. Illustrations of contactor configurations

See Fig. B.1.

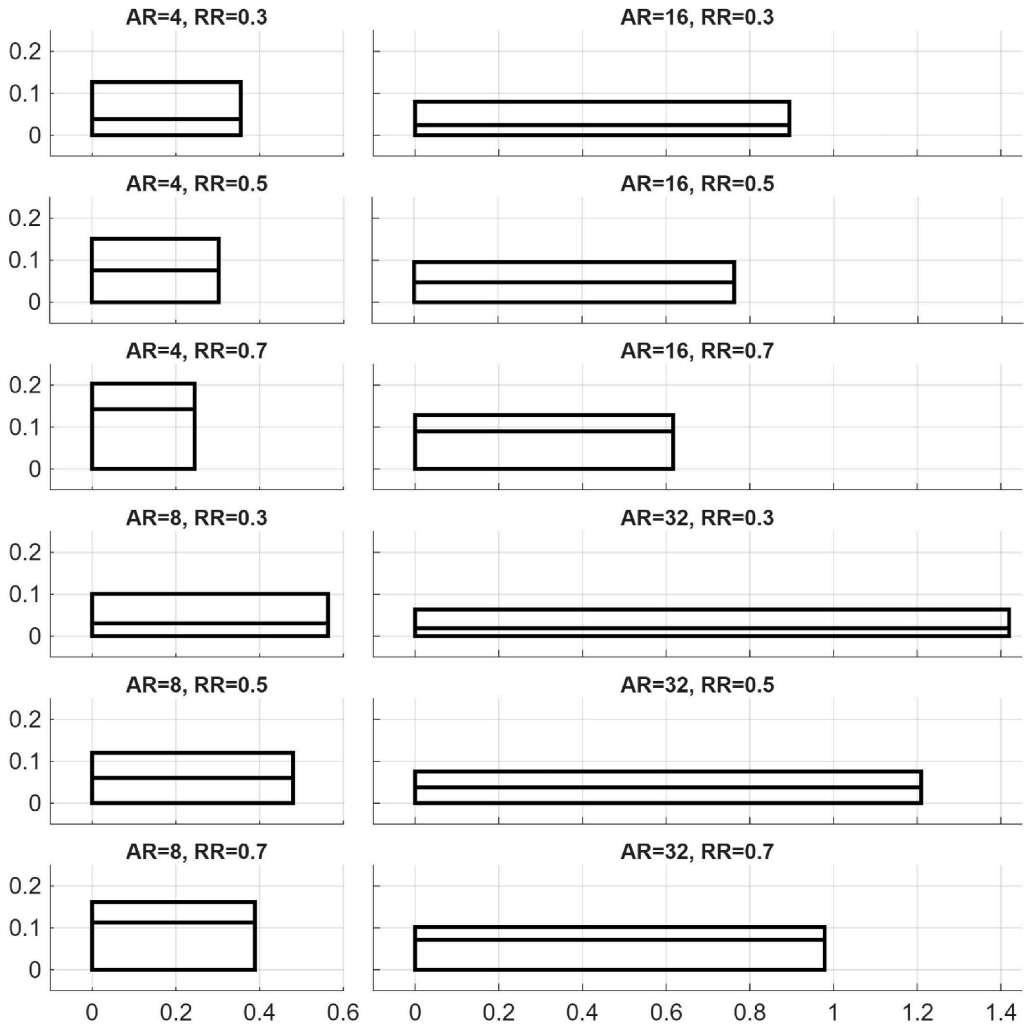



**Fig. B.1.** Geometry of each configuration.

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